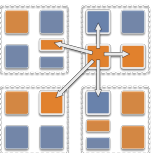


Exercise 4

- Particle program: Simulation of particles moving randomly on a 2D grid
- Goal: Calculate the average and maximum number of particles per chare, and print them periodically
 - But do not stop the simulation or synchronize to do this, use asynchronous methods to calculate and deliver this information



Exercise 4 details

- Grid has coordinates 100×100
- If the grid is $k \times m$ chares, each chare has a local portion of size $(100/k) \times (100/m)$
- With n particles, if a chare is in the x th row and y th column, it generates $k + (k(x+y)/(km))$ particles locally
 - This creates load imbalance
- In each iteration, randomly perturb each particle in a chare
 - If the new coordinates map to another chare, you will need to send that particle to the other chare
 - If the coordinate is out of bounds, revert to the previous coordinates

